ICMA CENTRE

IC103

Introductory Economics for Business and Finance

Lecture 4: Market Structures

Dr Ekaterina Glebkina

Henley Business School | University of Reading

Reading

Sloman, J., Wride, A. and Garratt,D (2018) “Economics” (10th edition).

https://ebookcentral.proquest.com/lib/reading/reader.action?docID=6824443#

- Section 7.1
- Section 7.2 up to page 195 (except Box 7.2)
- End of page 197 (benefits of perfect competition) and page 200
- Section 7.3 up to page 207 (except natural monopoly on page 202)
- Section 8.1 except Box 8.1
- Section 8.2 pages 221, 229-232 (from Non-collusive oligopoly), Box 8.3 optional

Section 1: Market Structures

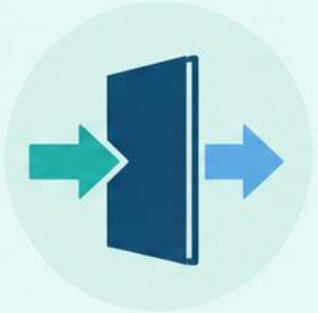
How competitive is a market?

Why Does Market Structure Matter?



Business Strategy

Pricing decisions
R&D and investment



Entry and Exit

Barriers to entering or leaving the market



Consumer Behaviour

How consumers respond to firms and prices



Regulation

How governments may regulate markets

Market structure affects how firms behave.

How Do We Classify Markets?

Classifying markets by **degree of competition**

1. How many firms are there?
2. How easy is it for firms to enter or leave the market?
3. Are products identical or differentiated?
4. How responsive is demand to price?

The Four Market Structures

There are **four main market structures** that we will study.

i. Perfect competition
ii. Monopoly
iii. Monopolistic competition
iv. Oligopoly

Image Source: https://gamerant.com/best-versions-of-monopoly/

Comparing Market Structures

Type of market	Number of firms	Freedom of entry	Nature of product	Examples	Implications for demand curve faced by firm
Perfect competition	Very many	Unrestricted	Homogeneous (undifferentiated)	Cabbages, carrots (approximately)	Horizontal. Firm is a price taker
Monopolistic competition	Many / several	Unrestricted	Differentiated	Builders, restaurants	Downward sloping, but relatively elastic
Oligopoly	Few	Restricted	1. Undifferentiated 2. Differentiated	1. Petrol 2. Cars, electrical appliances	Downward sloping, relatively inelastic (shape depends on reactions of rivals)
Monopoly	One	Restricted or completely blocked	Unique	Many prescription drugs, local water company, train operators over particular routes	Downward sloping: more inelastic than oligopoly. Firm has considerable control over price

This table summarises the main differences between the four market structures.

For instance, under perfect competition, there are many firms and products are identical.

Entry is relatively easy, so firms have very little market power.

Measuring Market Concentration: Concentration Ratios

Concentration ratio: the combined market share of the largest firms in an industry.

- Gives an indication of the **degree of competition** in a market.
- Example: the five-firm concentration ratio is the combined market share of the five largest firms.
- **Higher concentration ratio → more con centrated market → generally less competition.**
- Concentration can be affected by economies of scale and barriers to entry

Concentration Ratios: UK Manufacturing Example

Industry	5-firm	15-firm	50 firm
Food products	10.68	21.89	40.91
Textiles	12.88	24.04	37.13
Wearing apparel	11.01	16.09	19.52
Basic pharmaceuticals	52.52	70.48	89.81
Computers and electronics	15.70	29.04	44.89
Electrical	17.33	29.70	48.70
Motor vehicles	63.86	77.70	86.50
Furniture	9.41	17.76	33.2

Source: ONS Annual Business Survey

This table gives us a real example of concentration ratios in UK manufacturing.

For example, the five-firm concentration ratio for motor vehicles is much higher than for food products.

This means that the largest five firms control a much larger share of the motor-vehicle market.

So the motor-vehicle industry **is more concentrated.**

Mentimeter EM

Open Menti to edit

Replace this slide

Add Menti slides

A market has many firms, differentiated products and relatively easy entry. Which market structure best describes it?

menti.com

5951 3473

Perfect competition	Monopoly	Monopolistic competition	Oligopoly
0	0	0	0

Waiting for participants

Show correct answer

Section 2: Perfect Competition

A market where firms are price takers

Perfect Competition: Key Characteristics

Characteristics/Assumptions:

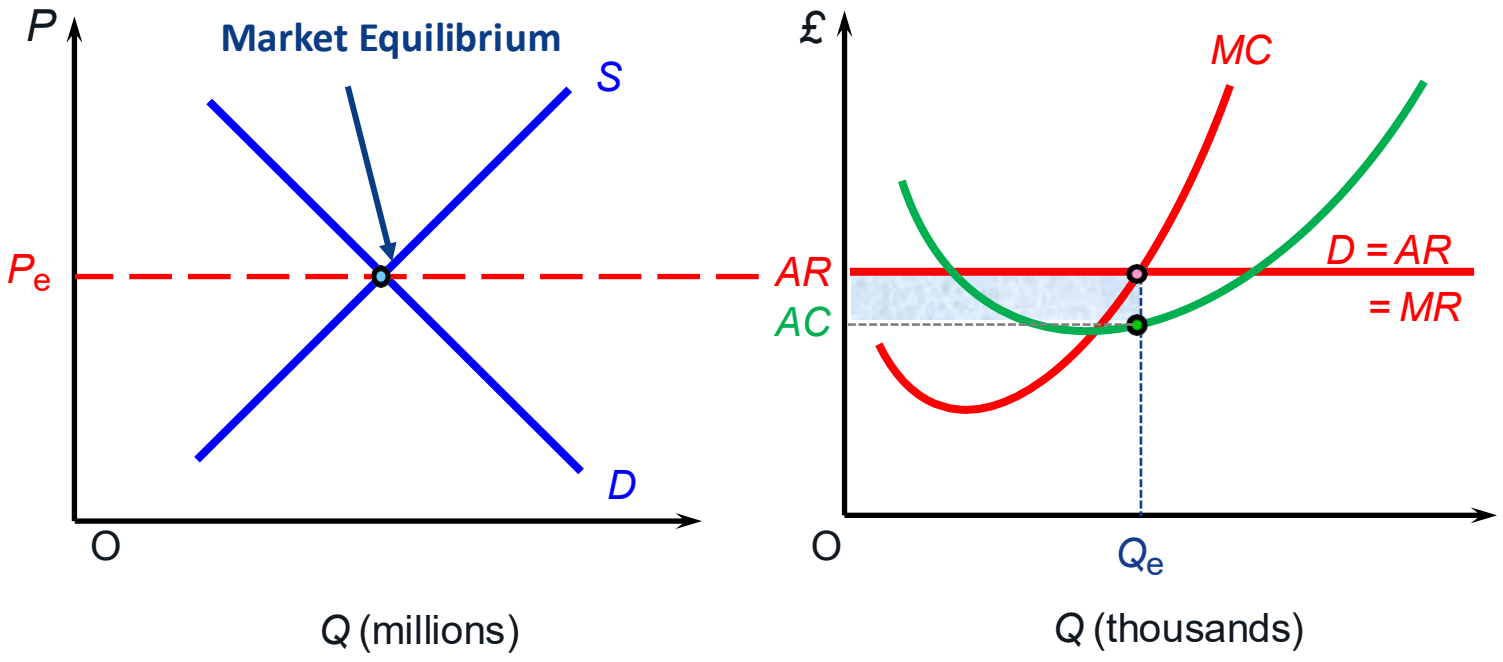
- **Large number of buyers and sellers**
- **Firms are small relative to the market**
- **Identical / homogeneous products**
- **No barriers to entry or exit**
- **Perfect information**
- Firms are **price takers**



Individual firm demand is perfectly elastic at the market price.

Perfect Competition: Market and Firm Equilibrium

Market: supply and demand determine the price



(a) Industry (whole market)

(b) Firm

Price taking firm: accepts market price

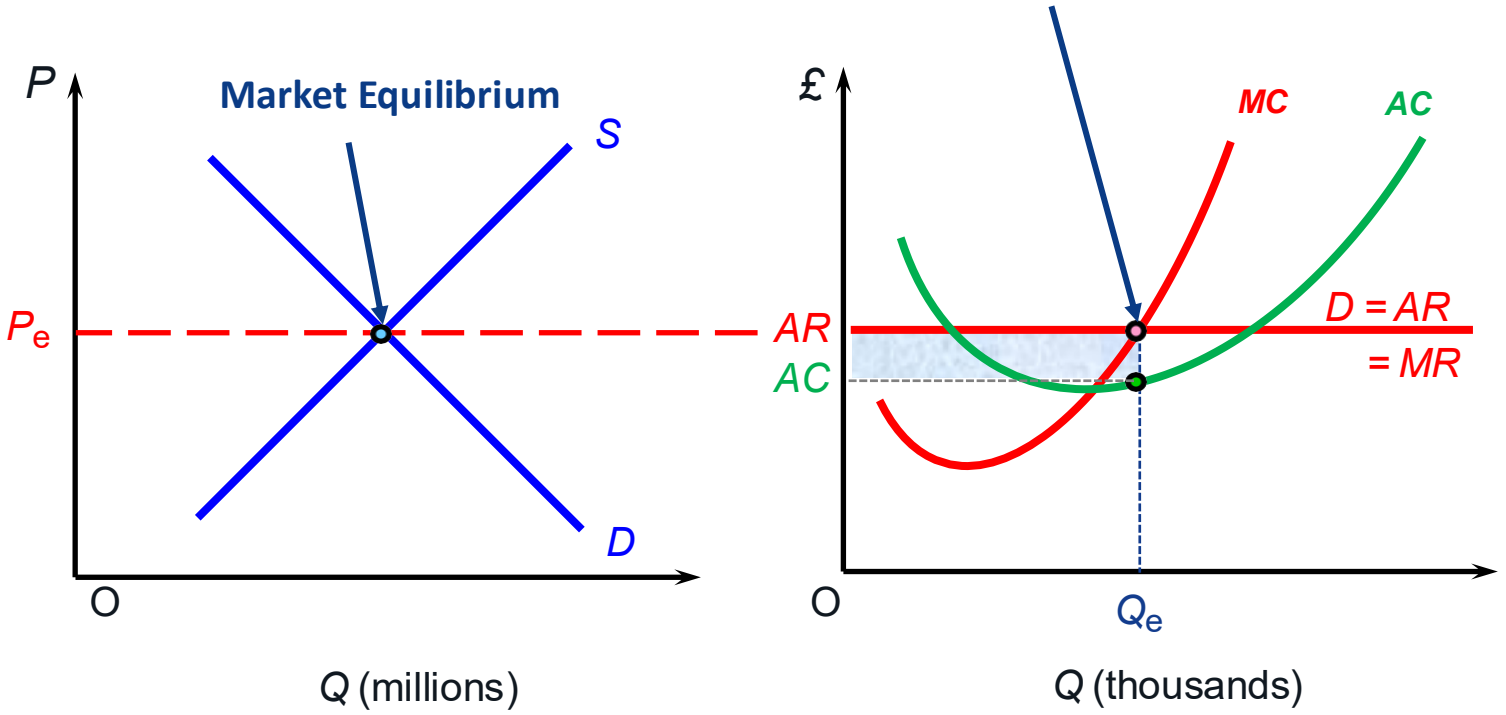
Horizontal demand curve that an individual firm faces.

Additional revenue for one more unit of product

$P_e = AR = MR$

Perfect Competition: Short-Run Profit Maximisation

Under perfect competition at the optimum **marginal revenue (MR) = marginal Cost (MC) = Price P_e**



(a) Industry (market)

(b) Firm

Stage 1: Find the profit-maximising output

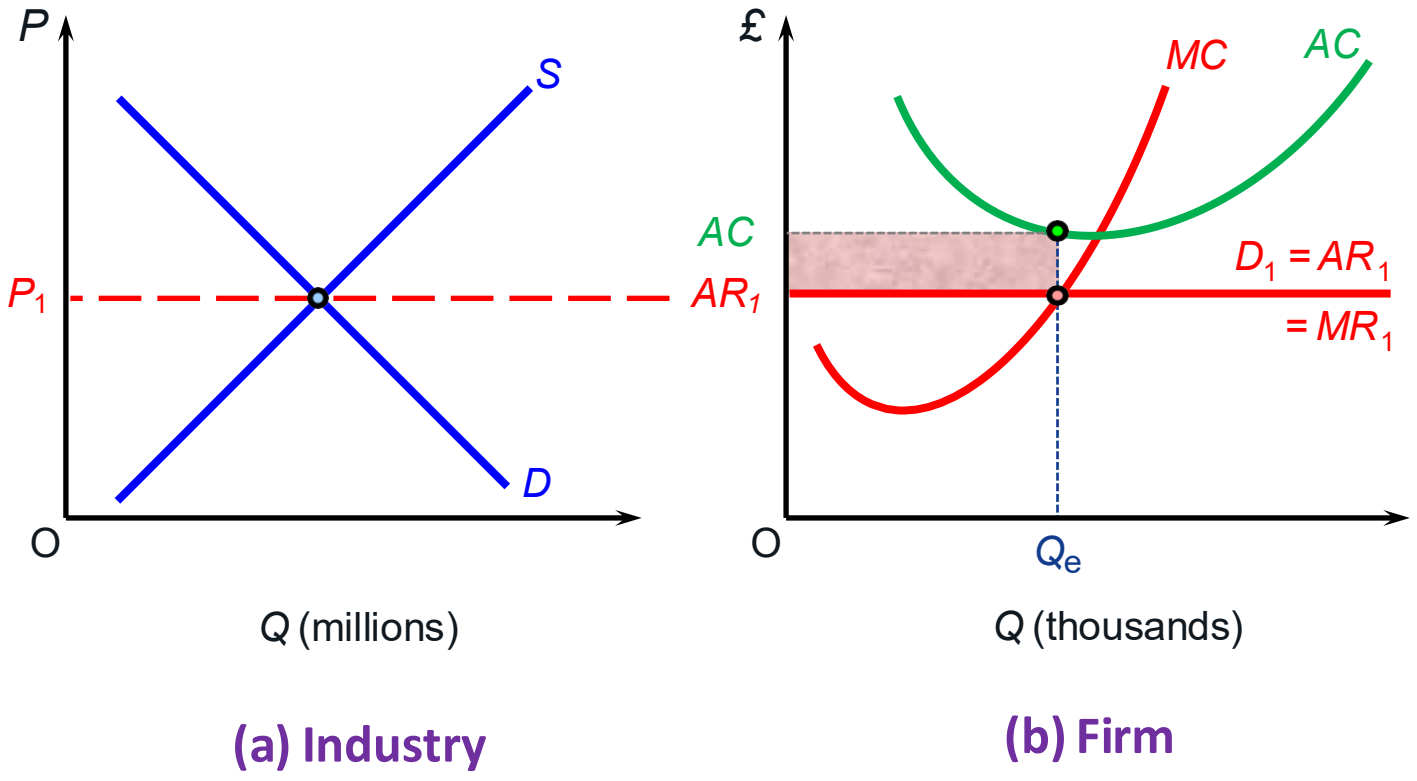
$$MR = MC$$

Stage 2: Measure profit

$$Profit = (AR - AC) \times Q_e$$

MR = Marginal Revenue
MC = Marginal Cost
AR = Average Revenue
AC = Average Cost

Perfect Competition: Short-Run Loss Minimisation



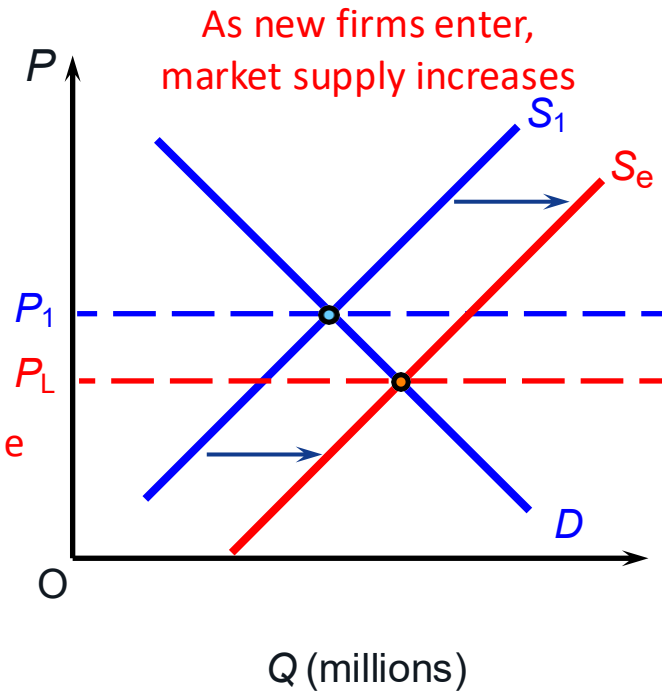
Loss is minimised where MC = MR.

Here, $\mathbf{AC} > \mathbf{AR}$, so the firm makes a loss.

The same $MR = MC$ rule applies whether the firm earns a profit or makes a loss.

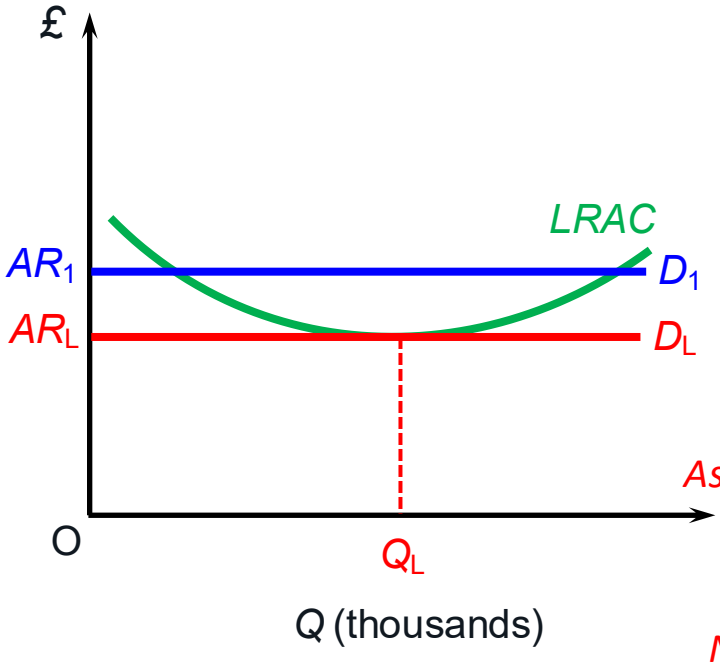
Perfect Competition: Entry and Long-Run Adjustment

In a perfectly competitive market,
Supernormal profit → new firms enter → market supply increases → price falls → profit falls



This pushes the market price down

(a) Industry



(b) Firm

Supernormal profits ($P1 > AC$)

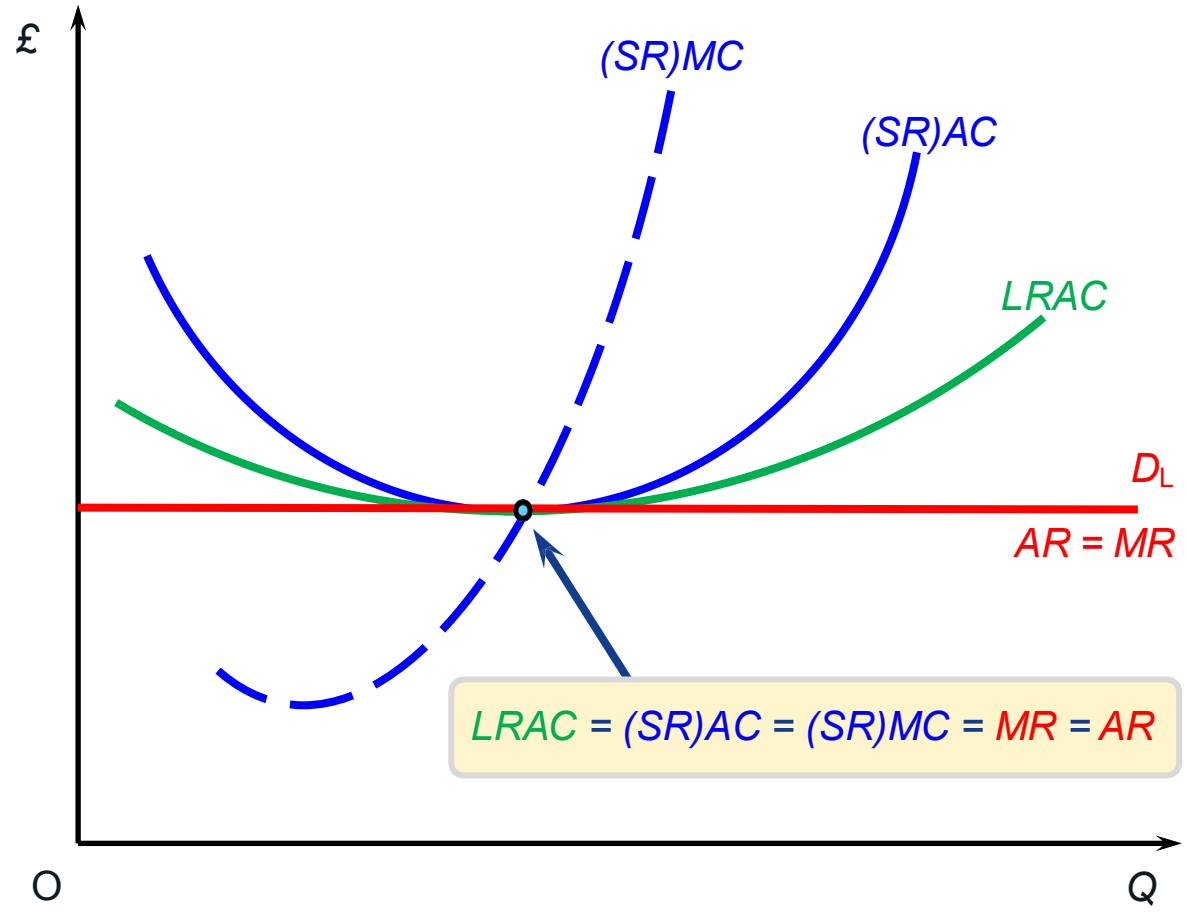
Profits return to normal

As the price falls, the profits of existing firms fall.

Entry continues until firms earn only normal profit.

Normal profit means that economic profit is 0 (firm earns enough revenue to cover all its costs, including the owner's opportunity cost.)

Perfect Competition: Long-Run Equilibrium



In the long run, firms earn **normal profit**
$TR = TC$ (economic profit $= 0$)

The firm still chooses its output where MR equals MC.

It is also producing at **minimum average cost.** This means it is producing at the lowest possible average cost.

In long-run equilibrium:
$P = AR = MR = MC = AC$

SRAC = Short-Run Average Cost
SRMC = Short-Run Marginal Cost
LRAC = Long-Run Average Cost

Perfect Competition: Advantages and Disadvantages

Advantages

1. Production at minimum cost
2. Competition encourages efficiency
   strong competition puts pressure on firms to be efficient
3. Little need for advertising
   little reason to spend large amounts on advertising when products are identical

Disadvantages

1. Less product variety
2. Lower profits may reduce investment
3. Less competition through product design and differentiation



Example that approximate perfect competition:

Agriculture market: many sellers, similar products, individual farmers cannot set the market price.

Section 3: Monopoly

A market with a single seller

Monopoly: Definition and Barriers to Entry

- Monopoly: a market structure characterized by a **single seller** in the market.
- Fundamental cause of monopoly: **barriers to entry**
  - structural/natural barriers - characteristics of the industry

    For example: starting the business may require extremely large investment

  - strategic/artificial barriers

    For example: exclusive contract - a firm signs agreements that stop rivals accessing key distributors

Barriers to entry protect the monopolist from competition.



Monopoly: Sources of Barriers to Entry

Economies of scale / natural

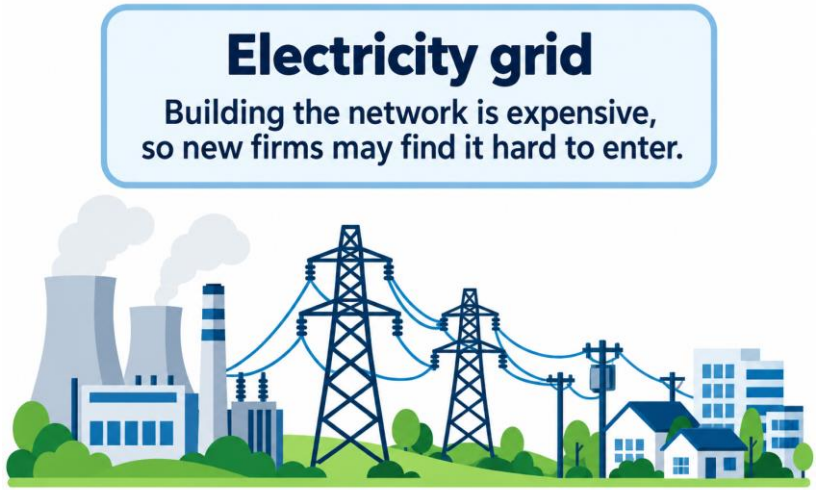
One large firm may produce at lower average cost than several smaller firms.

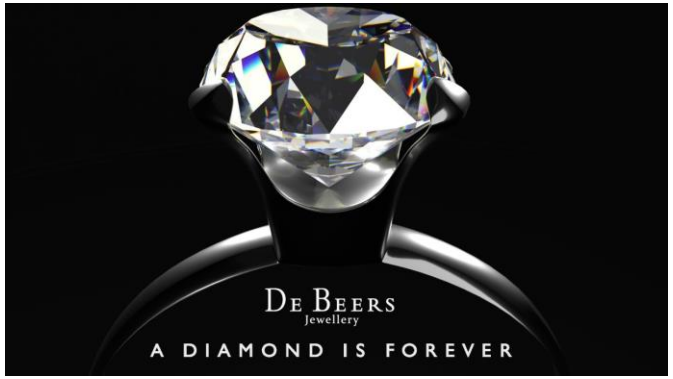
Control of key resources

A firm may control an essential input.

Exclusive contracts / legal

Other firms may be prevented from entering.



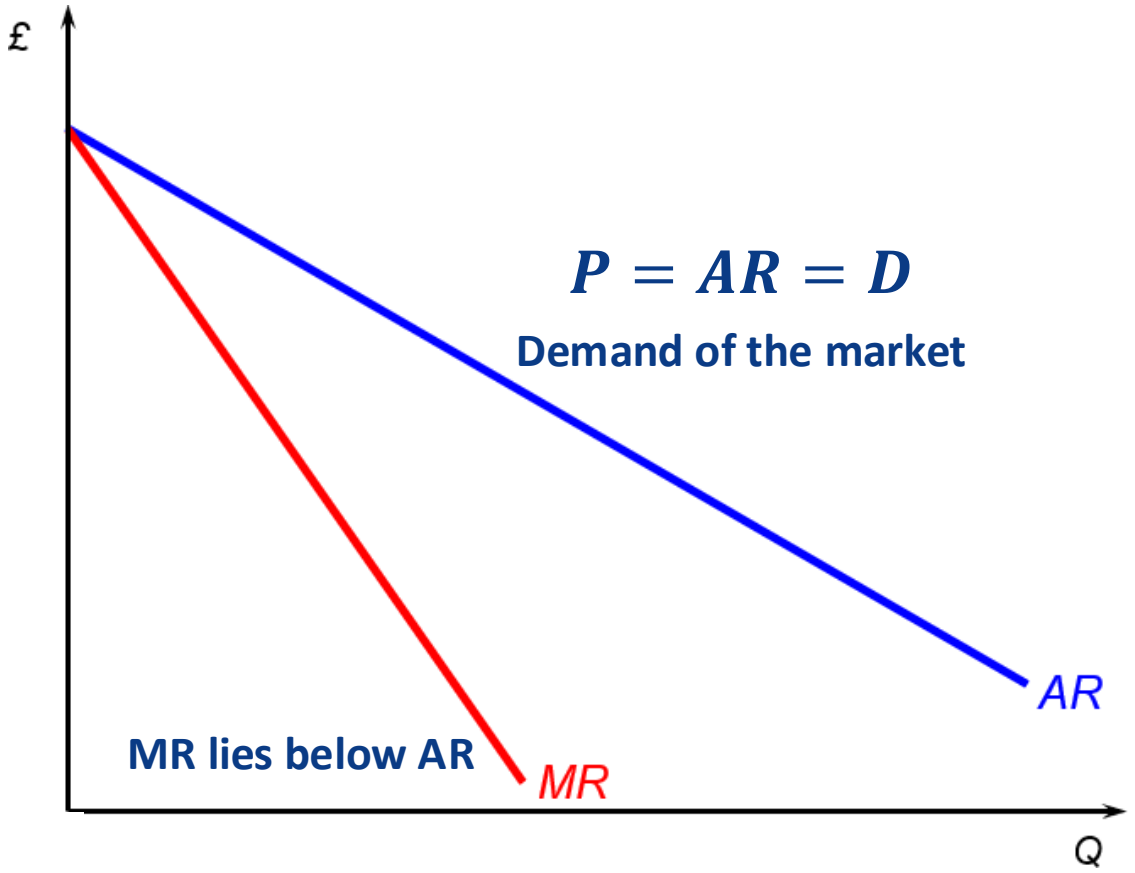


A historical example is De Beers and the supply of diamonds.
Control of a key resource can make entry difficult for rivals.

Intellectual Property Office

Patents and intellectual property rights can prevent competitors from producing the same product.

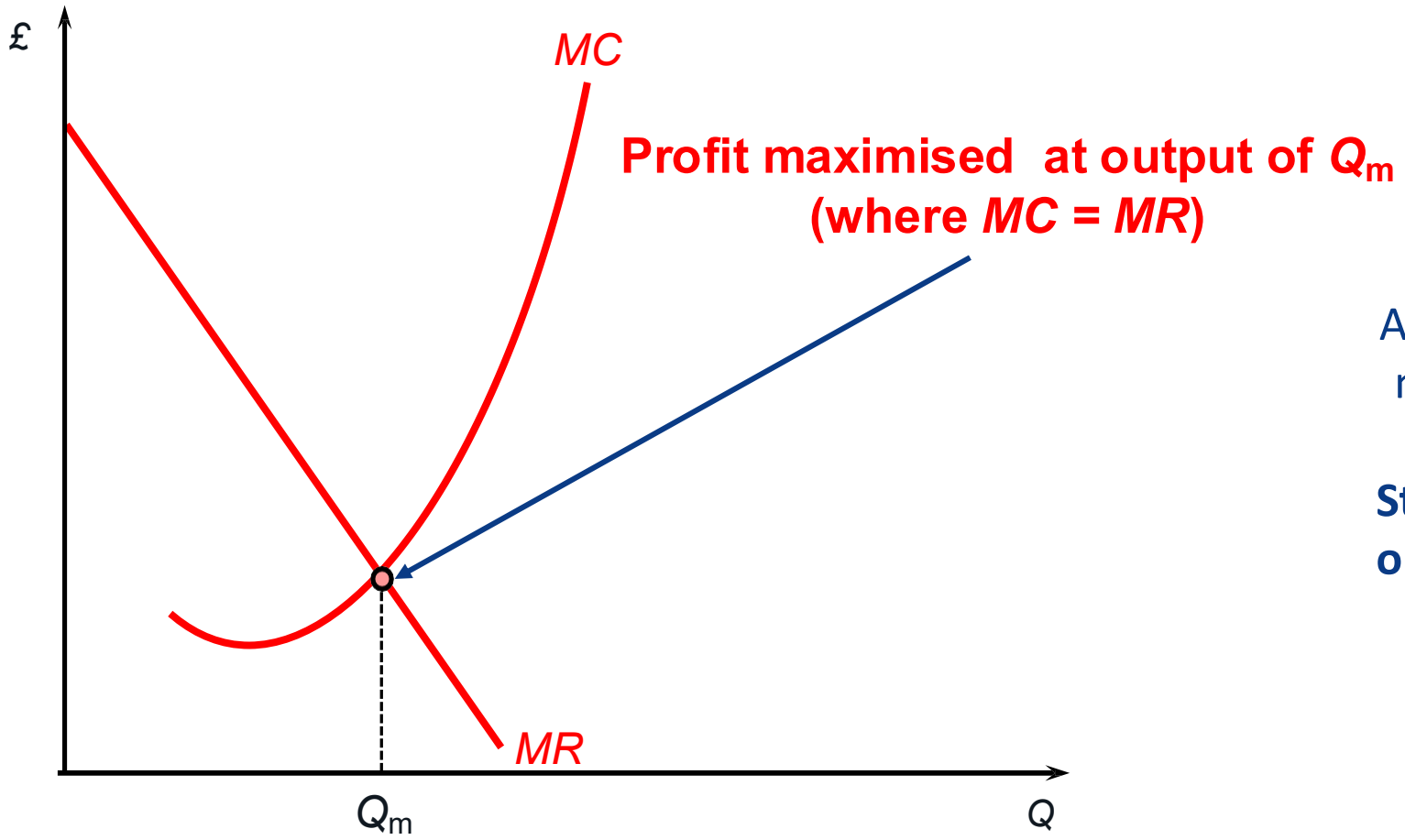
Monopoly: Average Revenue and Marginal Revenue



A monopoly faces a downward sloping demand curve, i.e. to sell more output, firm has to lower the price.

Marginal revenue lies below AR.
Because lowering the price to sell another unit also reduces the revenue received on the units that could previously have been sold at the higher price.

Monopoly: Profit-Maximising Output

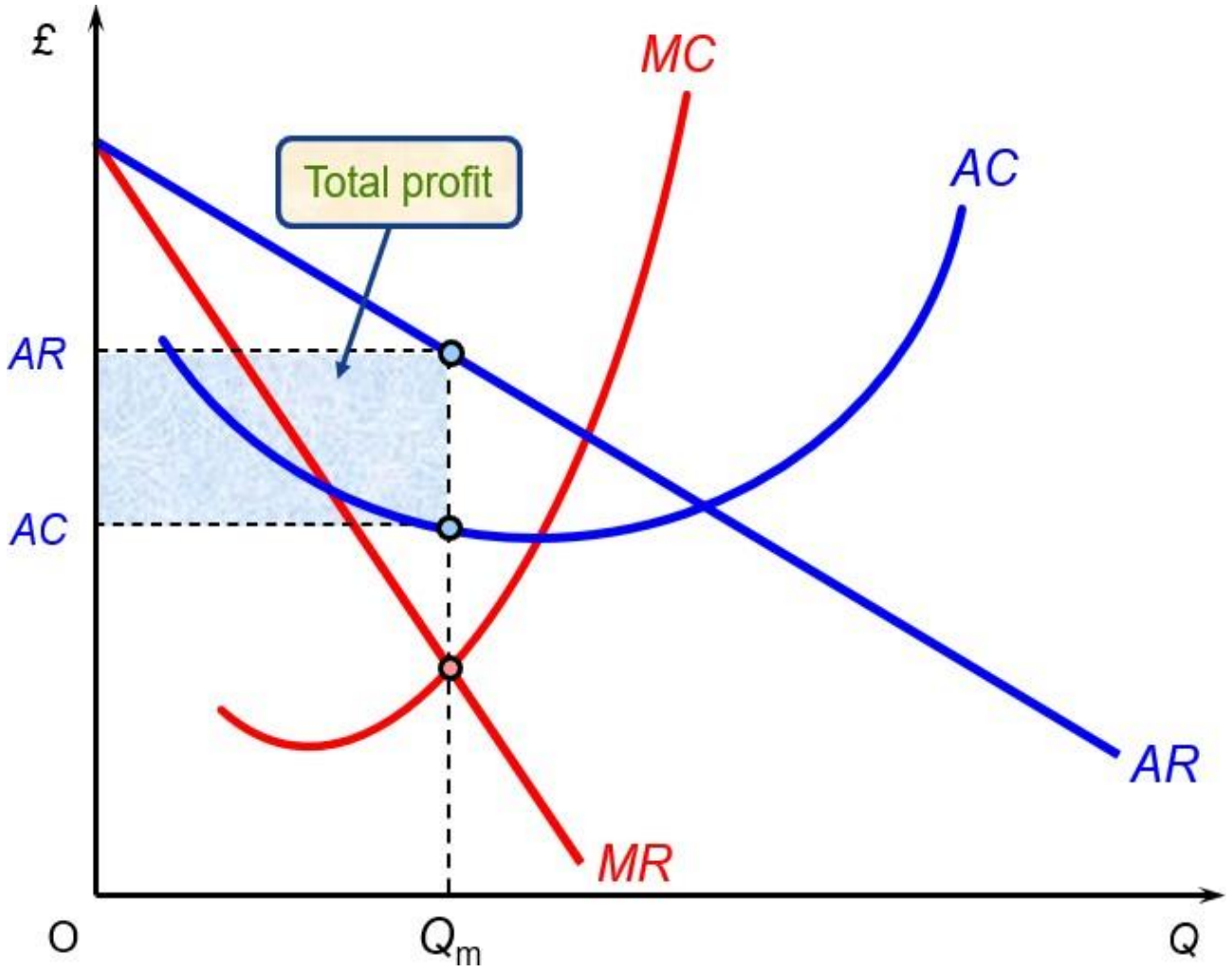


A monopolist follows the same profit-maximisation rule as any other firm:

Stage 1: Find the profit-maximising output

$MR = MC \rightarrow$ determines Q_m

Monopoly: Measuring Profit



Stage 2: Calculate profit

At Q_m:

$$\text{Profit per unit} = AR - AC$$

Then:

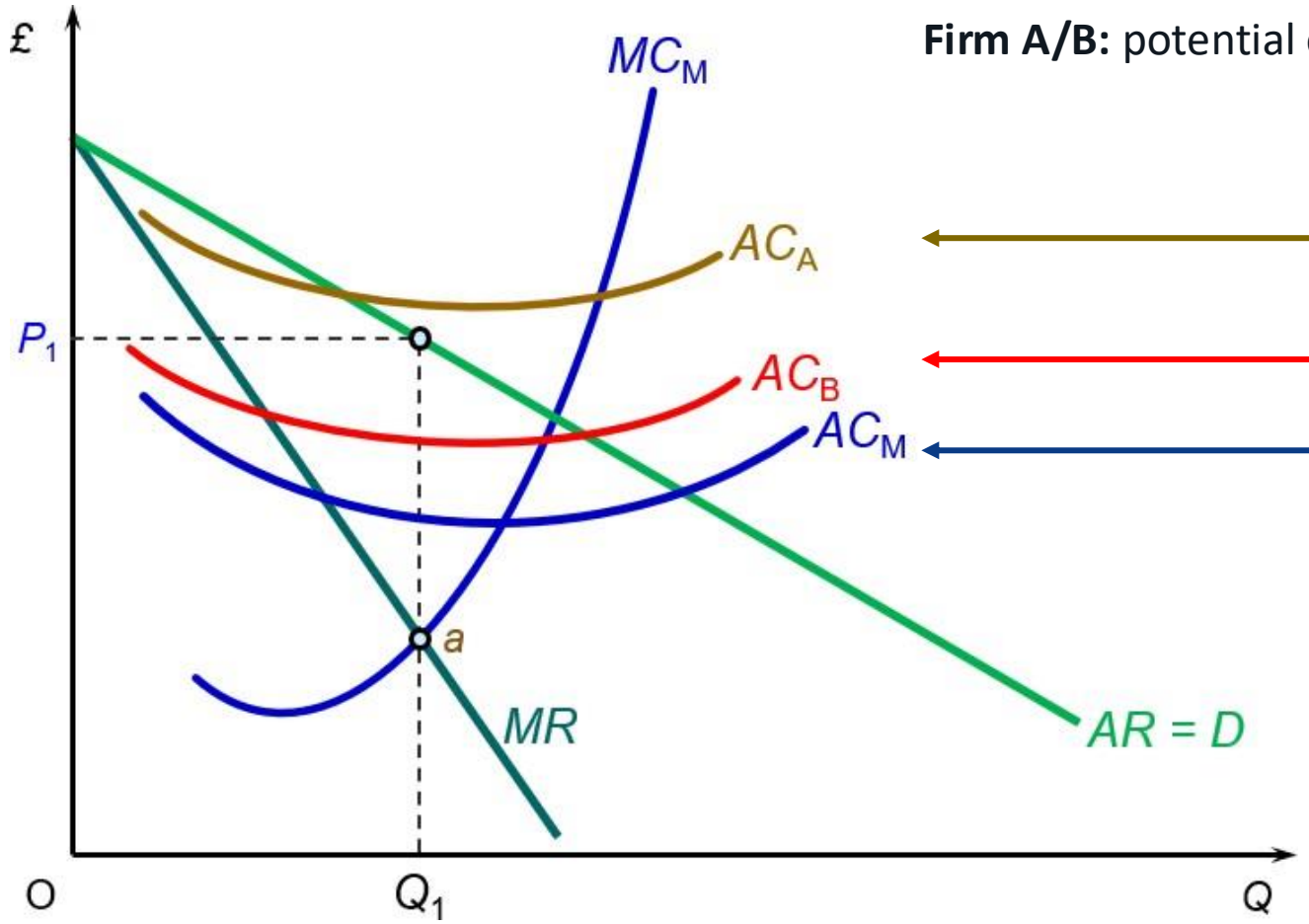
$$\text{Total profit} = (AR - AC) \times Q_m$$

The shaded rectangle on the graph represents total economic profit.

MR = Marginal Revenue
MC = Marginal Cost
AR = Average Revenue
AC = Average Cost

Monopoly: Potential Entry and Cost Advantage

Very large supernormal profits attracts new rivals to break into the industry.
Firm M: incumbent monopolist
Firm A/B: potential entrants



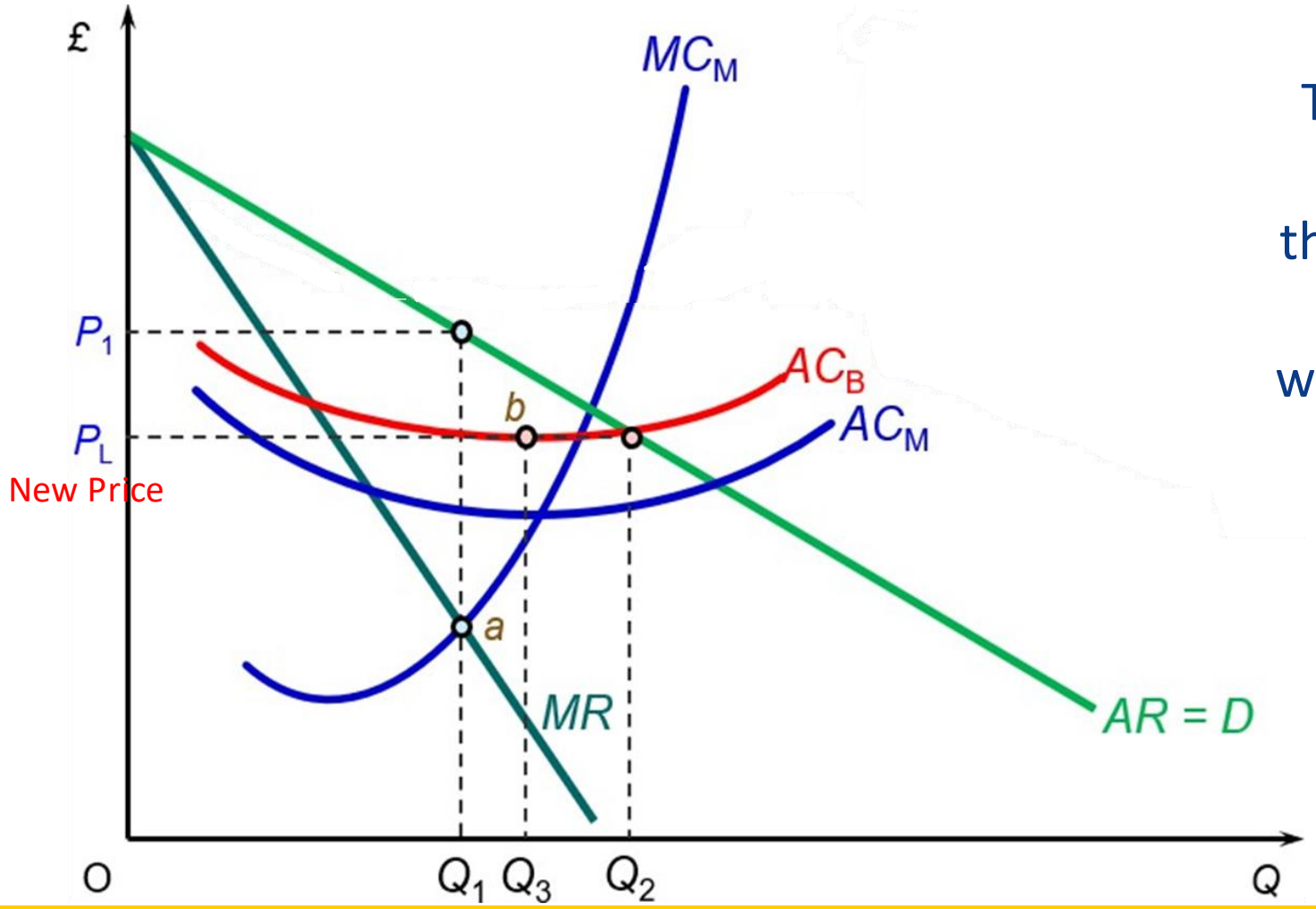
Firm A: $AC_A > P_1$, loss → does not enter

Firm B: $AC_B < P_1$, profit → may enter

Firm M: $AC_M < P_1$,
earns large supernormal profits due to its cost advantage (the lowest AC curve)

Price P_1= revenue per unit
Average cost AC = cost per unit

Monopoly: Deterring Entry Through Price Reduction



To prevent Firm B from entering the market,

the incumbent monopolist M reduces its prices to a level (from P_1 to P_L)

where Firm B earns only normal profit ($P_L = AC_B$)

Lower price can make entry less profitable for potential rivals.

Monopoly: Advantages and Disadvantages

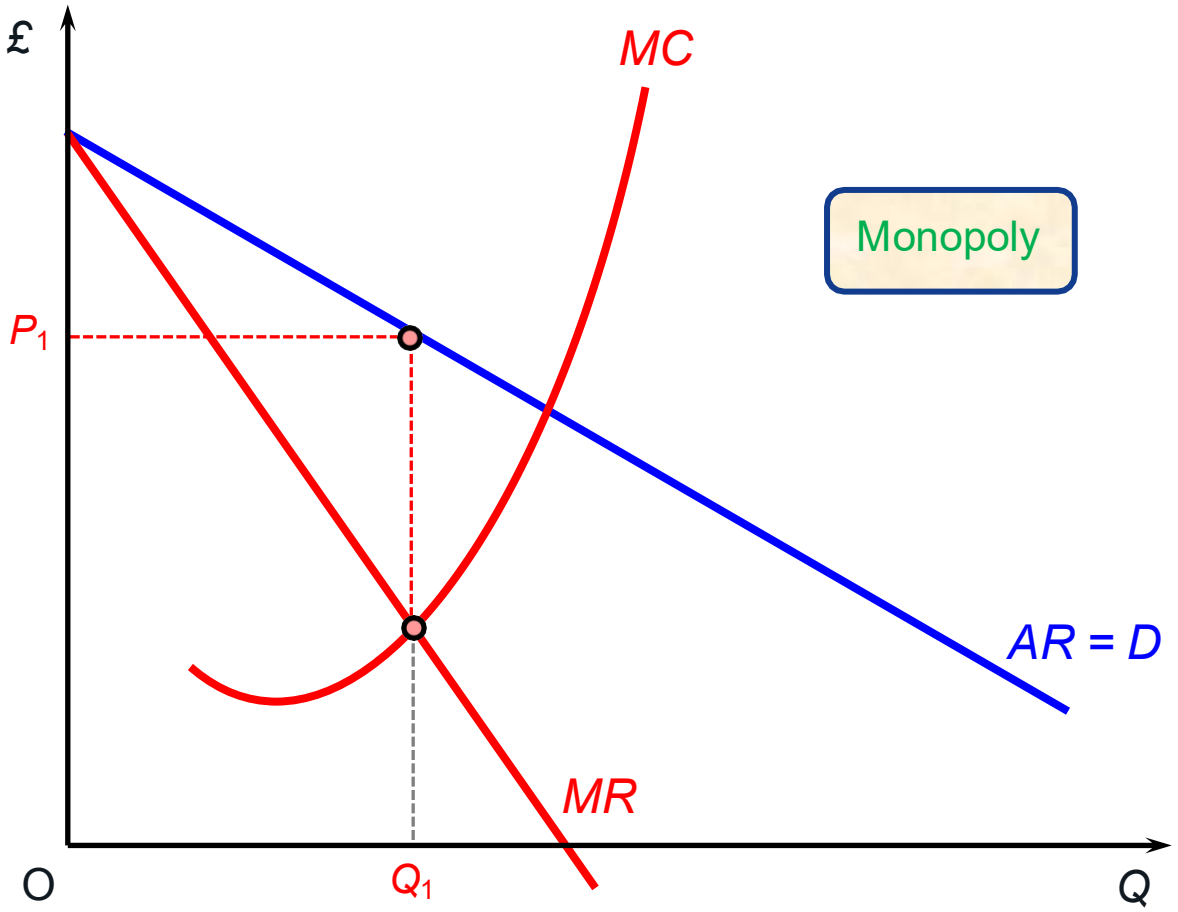
Disadvantages of monopoly

1. Higher prices
2. Lower output
3. Less competitive pressure
   the firm may become inefficient
4. Possible X-inefficiency
   not producing at the lowest possible cost
5. Potentially weaker incentives to innovate

Advantages of monopoly

1. Economies of scale
2. Large profits may finance investment
3. Competition for corporate control
   threat of takeover can pressure managers to improve efficiency

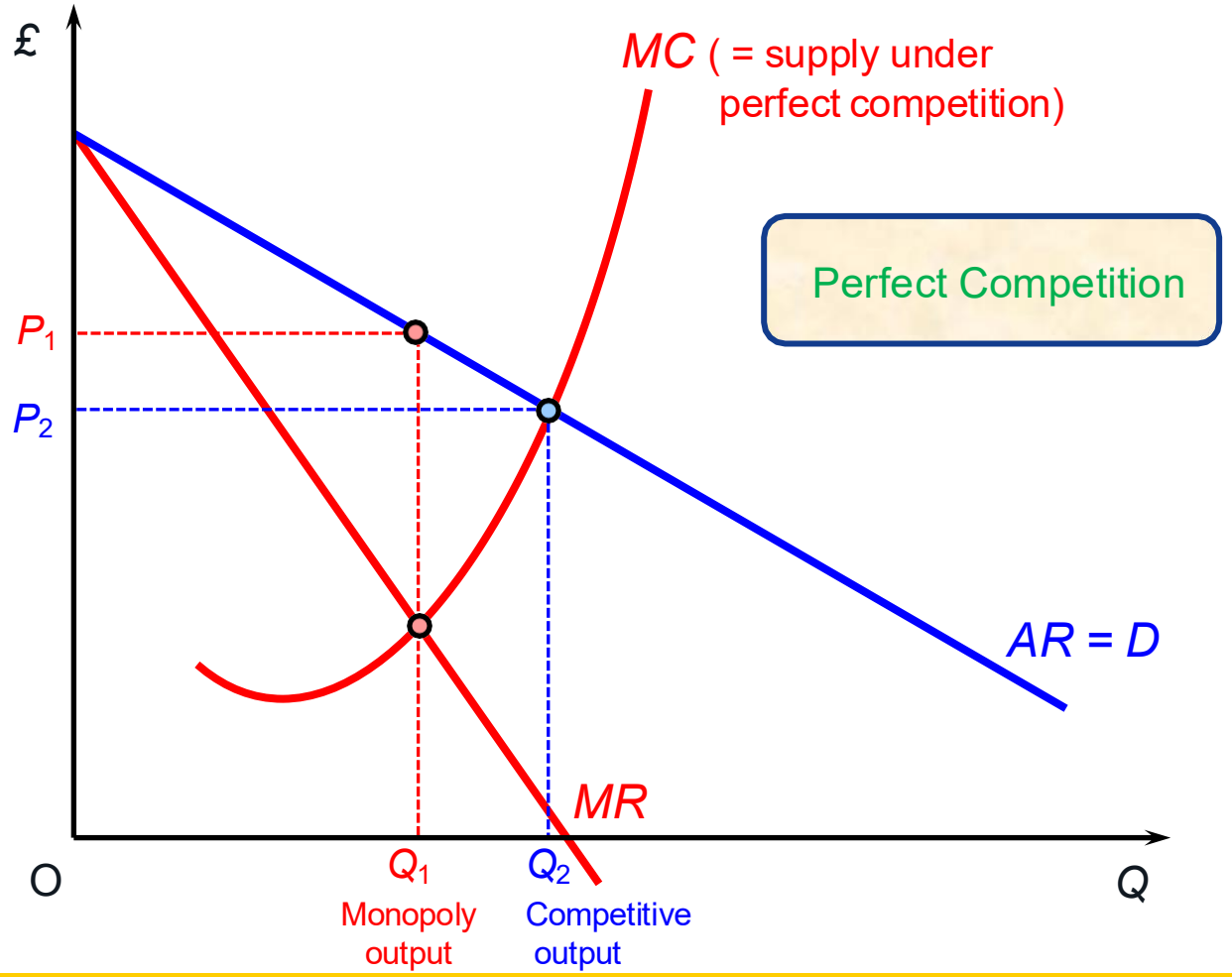
Monopoly vs Perfect Competition: Output Choice



Under monopoly, output is chosen where MR equals MC.

The monopoly output is Q_1.

Monopoly vs Perfect Competition: Price and Output



Under perfect competition, output is chosen where the market demand curve intersects the market supply curve, represented by MC: $D = MC$.

This gives a lower price P_2 and higher output Q_2.

Monopoly → higher price, lower output

Mentimeter EM

Open Menti to edit

Replace this slide

Add Menti slides

Compared with perfect competition, a profit-maximising monopoly will normally produce:

menti.com

5951 3473

0	0	0	0
More output at a lower price	More output at a higher price	Less output at a higher price	The same output at the same price

Waiting for participants

Show correct answer

Section 4: Monopolistic Competition

Many firms, but differentiated products

Monopolistic Competition: Key Characteristics

Assumptions of monopolistic competition

1. many or several firms
2. free entry
3. firms sell **differentiated product (similar but not identical)**

Example:
Restaurant / café
there are many restaurants and cafés, but each tries to be different through food quality, menu, location, atmosphere, branding, or service. Because products are differentiated, each firm has some control over its price.



Monopolistic Competition: Short-Run and Long-Run Equilibrium

Equilibrium of the firm

Short run

- $MR = MC$
- Supernormal profit possible

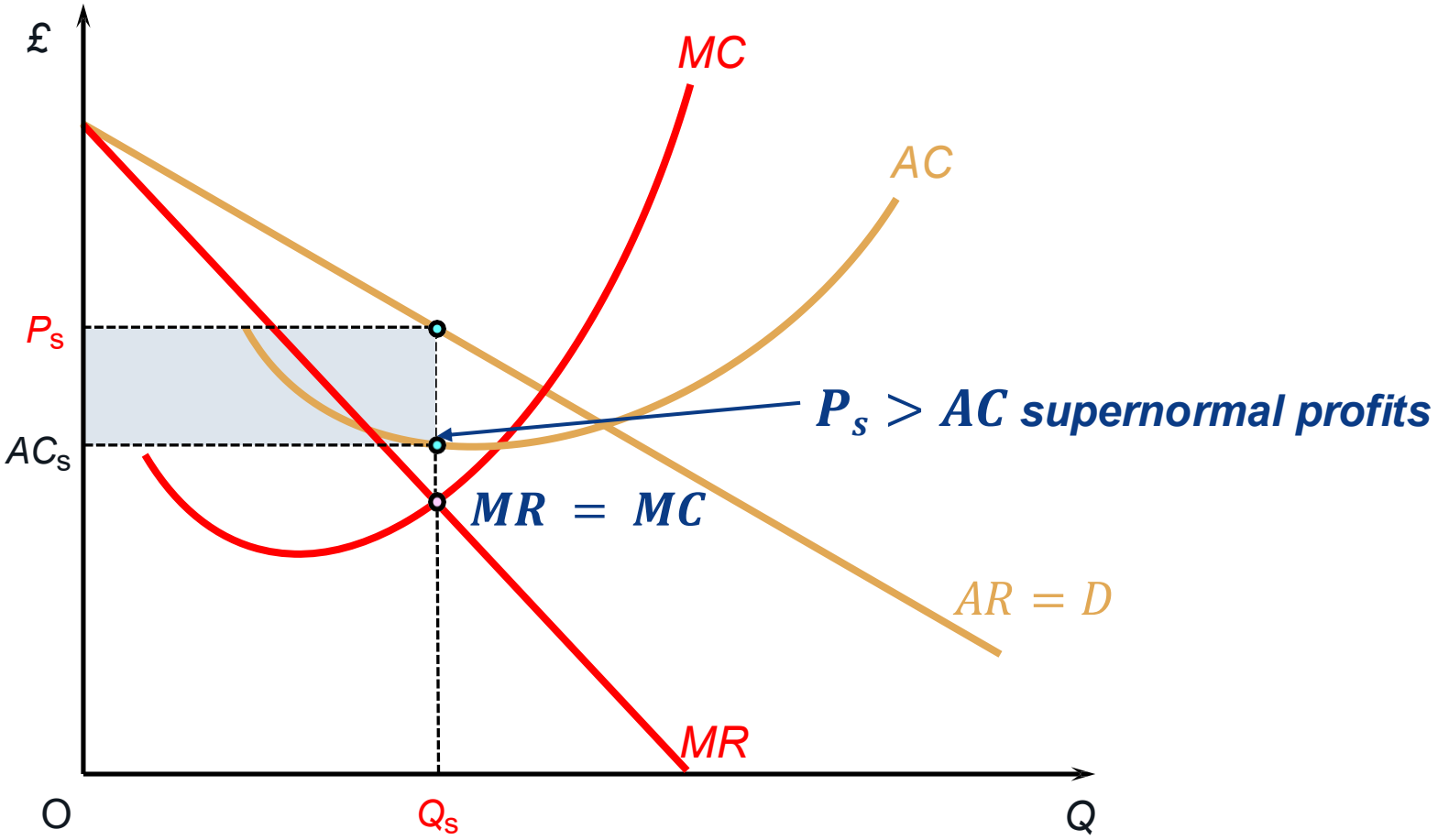
Because entry is relatively easy, high profits attract new firms.

Long run

- $MR = MC$
- $AR = AC$
- Normal profit

Competition reduces the demand faced by each existing firm. Eventually firms earn normal profit.

Monopolistic Competition: Short-Run Profit Maximisation



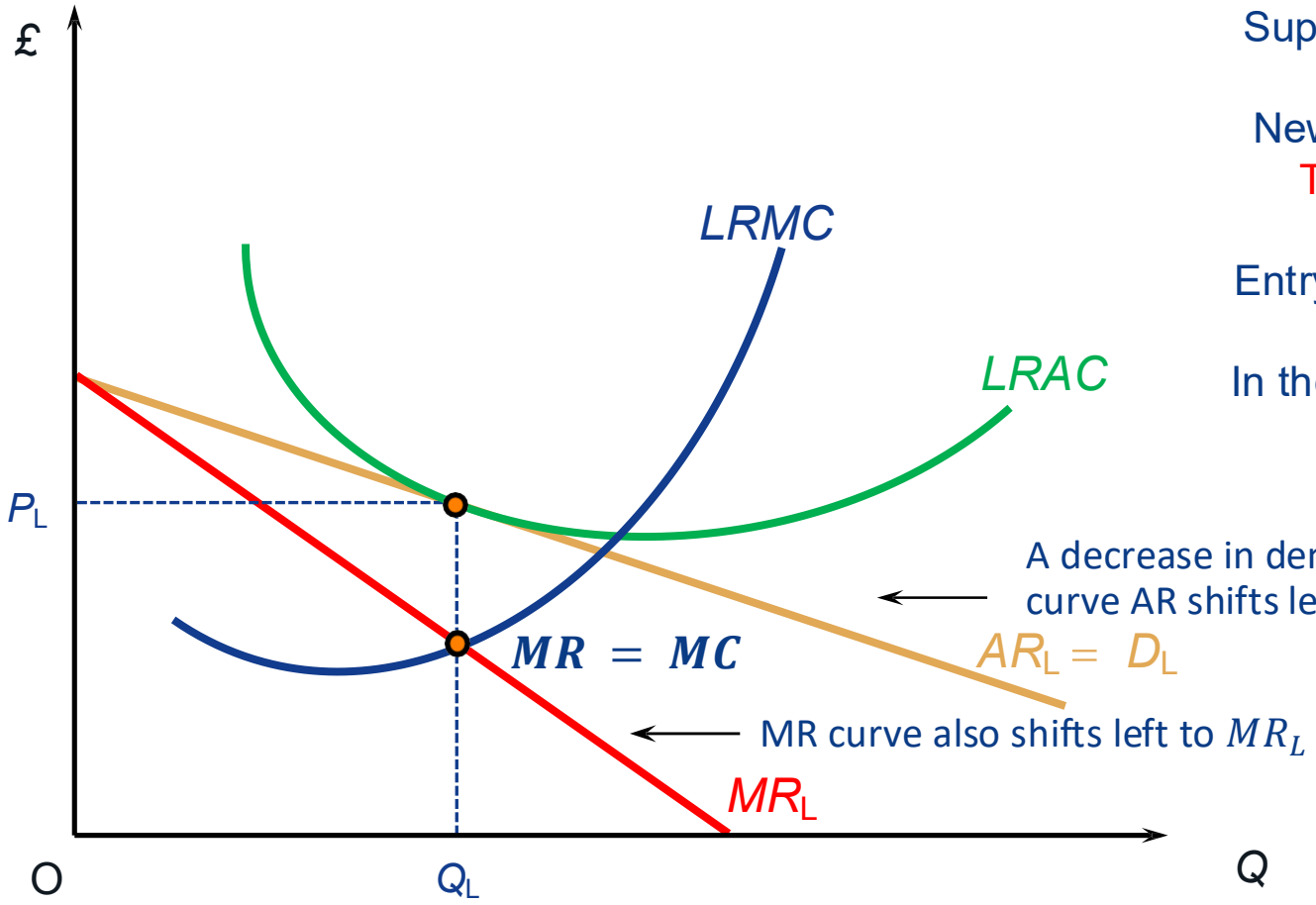
In the short run, the firm chooses output where MR equals MC.

Because the firm has a differentiated product, it faces a downward-sloping demand curve.

At this output, price may be above average cost.

If so, the firm earns supernormal profit.

Monopolistic Competition: Long-Run Equilibrium



Price falls to P_L

A decrease in demand causes the demand curve AR shifts left to $AR_L = D_L$

MR curve also shifts left to MR_L

Supernormal profits attract new firms because entry is relatively easy.

New firms give consumers more alternatives.

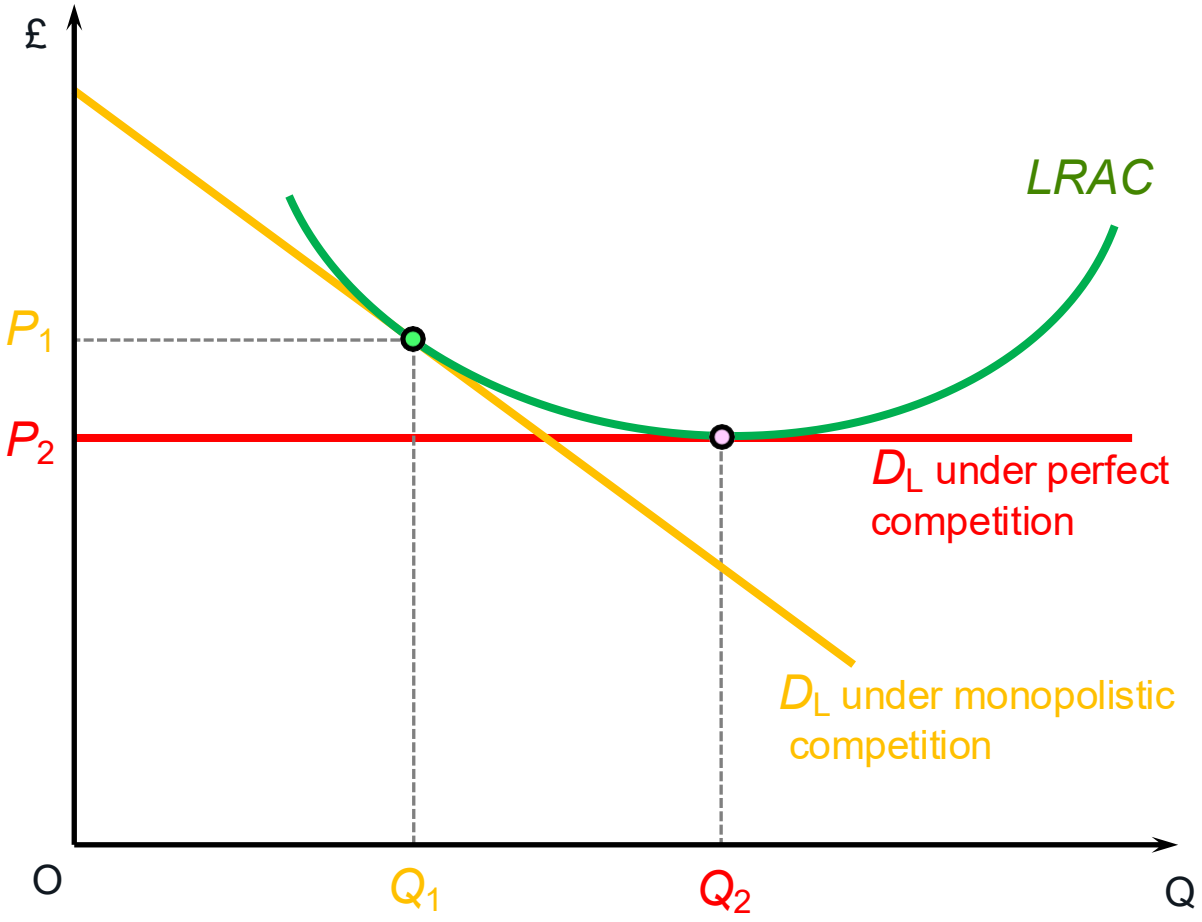
This reduces the demand faced by each individual firm.

Entry continues until the firm earns only normal profit.

In the long run, AR equals AC at the equilibrium output.

Supernormal profit → entry → demand for each existing firm falls → profit falls → normal profit

Monopolistic Competition vs Perfect Competition



Monopolistic competition

- Higher price
- Lower output
- Excess capacity= the firm produces below the output level Q_2 that gives the lowest average cost.

Monopolistic Competition: Non-Price Competition

Firms do not compete only through price.

Product Development
Better features, design or innovation

Branding & Advertising
Make the product stand out

Customer Service
Create a better customer experience

Quality
Improve the product or service

Section 5: Imperfect Competition — Oligopoly

A market dominated by a few large firms

Oligopoly: Key Characteristics

- Oligopoly occurs when just **a few firms** between them share a large proportion of the industry.
- The products offered are similar or identical products.
- Key features of oligopoly
  - High barriers to entry
  - Firms are interdependent
    Because there are only a few firms,
    each firm's decisions affect its competitors
  - Firms may compete or collude

OPEC = Organization of the Petroleum Exporting Countries

Oligopoly: Non-Collusive Models

- **Non-collusive oligopoly:** firms do **not** make an agreement with each other — formal, informal or tacit.

- Key issue:
  Each firm must think about how its rivals may react to its decisions.

- Three well-known models:
  - Cournot – firms compete by choosing **quantity**
  - Bertrand – firms compete by choosing **price**
  - Kinked demand curve – firms consider **how rivals may respond** to price changes

Oligopoly: The Cournot Model

The Cournot model of duopoly (a market with two firms)

- Assumptions
  - Firms choose a **quantity (output)** without knowing exactly how much the other firm will produce
  - Production is relatively inflexible / **has long lead times** (difficult to change quickly)
  - **Market price adjusts** to total output produced by both firms
  - Firms sell **homogeneous products**



Cournot = competition in quantity

Oligopoly: The Bertrand Model

The Bertrand model of duopoly

- Assumptions
  - Firms choose **prices** without knowing exactly what price its rival will choose
  - Prices set in advance – **cannot easily be changed**
  - But firms can **adjust output quickly** – no capacity constraints
  - Firms sell **homogeneous products**.

Bertrand = competition in price

Oligopoly: The Kinked Demand Curve

This model focuses on how competitors react to changes in price.

- Assumptions
  - If a firm **cuts its price**, rivals are expected to **follow**.
  - If a firm **raises its price**, rivals are expected **not to follow.**



Why? If they do not follow the price increase, they may attract some of that firm's customers.

  - This creates a **kink** in the firm's demand curve.
  - The model helps explain why prices in oligopoly may be relatively stable.

Rivals react differently to price cuts and price increases.

Mentimeter EM

Open Menti to edit

Replace this slide

Add Menti slides

Which statement correctly matches the oligopoly model with the firm's main strategic choice?

menti.com

5951 3473

0	0	0	0
Cournot → price; Bertrand → quantity	Cournot → quantity; Bertrand → price	Cournot → advertising; Bertrand → quantity	Both models assume firms choose only price

Waiting for participants

Show correct answer

References

- Sloman, J., Wride, A. and Garratt,D (2018) “Economics” (10th edition)
- Gregory, Mankiw, N. (2017) “Economics” Cengage Textbooks